STAFF REVIEW · H446 147 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A denial-of-service attack, potentially distributed (DDoS).

2. Referencing a candidate key in another relation.

3. RAM.

4. The internet layer.

I Do — source worked steps

1. Client web request — Identify the facts and make an initial prediction.

2. Proxy and firewall controls — Explain the mechanism or trace the intermediate stage.

3. HTTPS encrypted transfer — A managed web proxy can filter/log permitted requests; a firewall can restrict traffic flows; HTTPS/TLS encrypts the connection so local eavesdroppers cannot read credentials. Policy, MFA and endpoint protection address further threats.

We Do — guided CHECK

1. Traffic allowed on port 443
Identify the facts and make an initial prediction.
Malicious content reaches user
Explain the mechanism or trace the intermediate stage.
Endpoint controls matter
An allowed connection or email can carry malicious content. Endpoint malware detection, attachment scanning, patching, permissions and user training complement network filtering. A proxy may help inspect or block risky destinations.

You Do — source mark schemes

1. Firewall controls traffic according to rule/state [2]; proxy intermediates requests/cache/filter [2]; TLS protects communication in transit [2].

2. Encryption protects data in transit [2]; a malicious website may still use encryption [2]; verify destination, credentials or content separately [2].

3. Credit four linked controls: firewall, proxy/web filtering, VPN or TLS, MFA, patching/endpoint protection, least privilege or monitoring [2 each].

Exit Ticket — source answer

1. False. Proxy intermediates requests; a firewall applies traffic access controls.